
CSC 220

MAPS AND HASH MAPS

Fahmida Hamid

CASE 01: STUDENT RECORDS

Student ID	Score
S101	85
S102	92
S103	78
S104	92

- How do we store the data?
- Lists?
 - *Any disadvantages?*

USE CASES

- A **university course registration system** with millions of enrollment records across many years.
- The internet with millions of pages and growing...

MAP: THE ABSTRACT IDEA

- A **Map** is a conceptual blueprint for storing data.
 - It is a collection of **key-value pairs**, where every unique key links to a specific piece of information.

S101 → 85

S102 → 92

S103 → 78

MAP: THE ABSTRACT IDEA

- A **map** directly associates each key with a value and
- typically lets us look up a value in **constant time**.

HASHMAP

A SPECIFIC, HIGH-PERFORMANCE
IMPLEMENTATION OF MAP

MAP & HASHMAP IN JAVA

- Map is an **interface**: it specifies operations such as **put** and **get**.
- HashMap is a class that implements those operations.
 - <https://docs.oracle.com/javase/8/docs/api/java/util/HashMap.html>

HASH

- What does Hash mean?
 - A HashMap does not normally search every entry from beginning to end.
 - It uses the key's **hash code** to narrow down where to look.
 - This makes lookup typically fast.

HASH

- `"S102"` → `hashCode()` → choose a location → find the entry → 92
 - Hash codes do **not** have to be unique.
 - Two different keys can lead to the same location. That is a **collision**. The map then checks which key actually matches using `equals()`.

Key A —┐

| — same location → compare keys with `equals()`

Key B —┐

HASH

- A **collision** does not mean that one key overwrites the other.
- A HashMap can retain both distinct keys.

HASHMAP

- A HashMap stores values under keys.
- When we call `get(key)`, it uses the key's `hashCode()` to narrow down where to search, then uses `equals()` to identify the matching key.
 - Two different keys can have the same hash code, so a matching hash code alone does not mean the keys are equal.
- If two keys are equal according to `equals()`, they **must** have the same hash code.

HASH FUNCTION

HASH FUNCTION

- A **hash function** takes a key and produces a number that helps us decide where to store or look for it.
- $h(x) = x \bmod 5$

Key (x)	(x mod 5)	Bucket
12	2	2
27	2	2
31	1	1
45	0	0

$$h(x) = x \bmod 5$$

Resolve collision?

NEXT TOPIC...
